

UNIVERSITY OF FRONTIER TECHNOLOGY, BANGLADESH
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PHISHSHIELD

PHISHING EMAIL DETECTION SYSTEM

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What is Phishing?

- Phishing is a **cyber attack** using fake emails to deceive users.
- Attackers try to steal **sensitive information** like passwords and financial data.
- Users often **cannot identify** malicious emails from legitimate ones.



3.4B

Phishing emails/day

36%

Of all data breaches

Existing Problems



Fake Sender

Spoofted email addresses
impersonating trusted sources



Malicious Links

URLs leading to fraudulent or
malware-infected sites



Urgent Scams

Messages creating false
urgency to pressure users



Identity Theft

Stealing personal credentials
for fraudulent purposes



Financial Fraud

Tricking victims into
transferring money or data

Project Overview

PhishShield is a rule-based email security system that analyzes multiple components of an email and determines whether it is safe or phishing.



Sender

Verifies the email sender's domain against a trusted list



Subject

Scans subject lines for suspicious keywords and patterns



Content

Analyzes email body text for phishing indicators

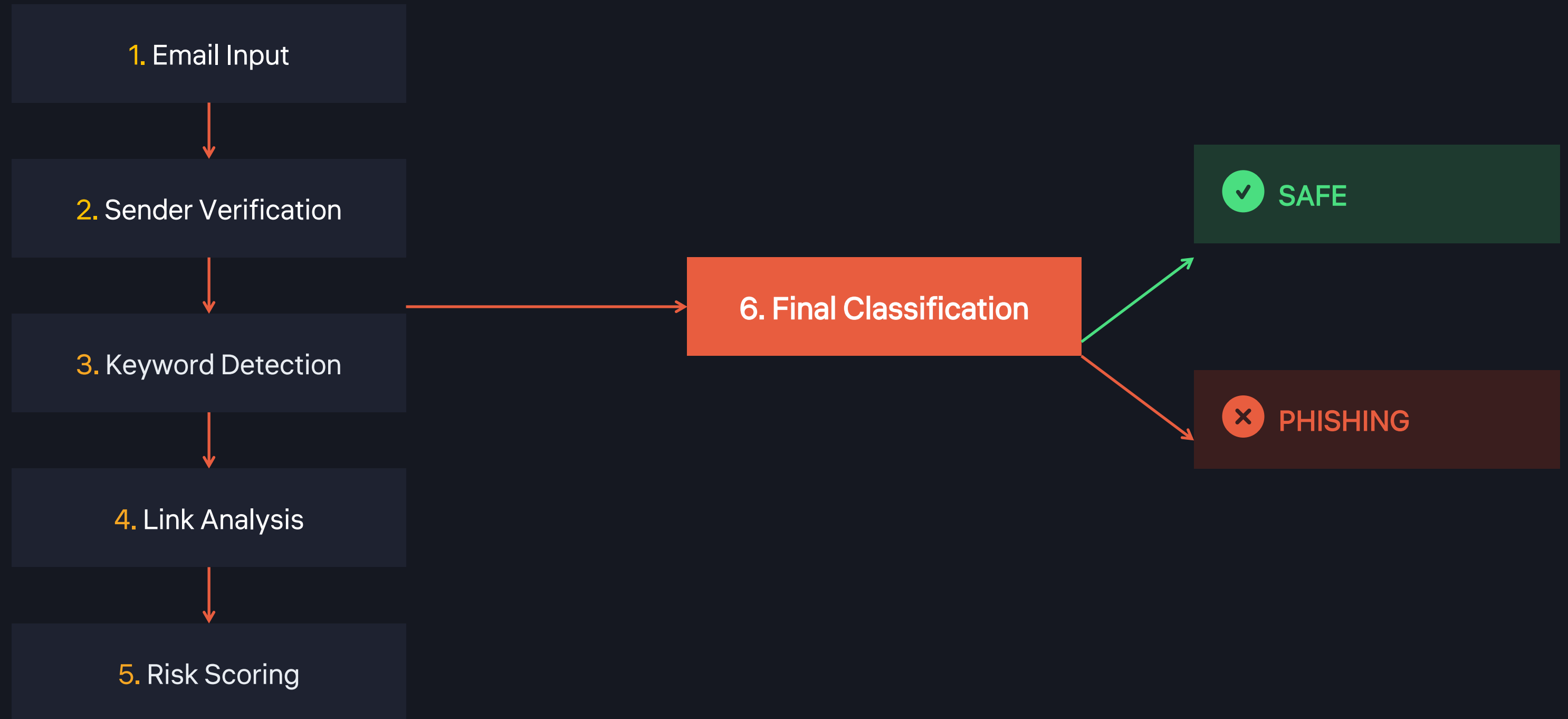


Links

Checks embedded URLs for HTTPS and short URL risks

Result: Determines whether an email is **SAFE** or **PHISHING**

System Workflow



Class Structure — Email Class

ATTRIBUTES

@ **Sender** — email source address

H **Subject** — email subject line

≡ **Content** — email body text

🔗 **Link** — embedded URL

METHODS

→ **getSender()** — returns sender address

→ **getSubject()** — returns subject line

→ **getContent()** — returns body content

→ **getLink()** — returns embedded URL

Sender Verification Module

TRUSTED DOMAINS

- ✓ @uftb.edu.bd
- ✓ @gmail.com
- ✓ @yahoo.com

DETECTION LOGIC

Unknown Domain



Add Risk Score



Generate Warning



Keyword Detection Module

SUSPICIOUS KEYWORDS

Lottery

Urgent

Verify Password

Click Here

Free Reward

Bank Account

IF FOUND



Increase Risk Score

+2 points per keyword match

HOW IT WORKS

1. Scan email subject and body text
2. Match against keyword database
3. Increment risk score for each hit


```

bool isTrusted = false;
for (string domain : safeDomains) {
    if (email.getSender().find(domain) != string::npos) {
        isTrusted = true;
        break;
    }
}
if (!isTrusted) {
    phishingScore += 2;
    flags.push_back("Suspicious or Unverified Sender Domain");
}
for (string keyword : suspiciousKeywords) {
    if (email.getSubject().find(keyword) != string::npos || email.getContent().find(keyword) != string::npos)
        phishingScore += 2;
    flags.push_back("Contains Phishing Keyword: '" + keyword + "'");
}
}

```

Output:

```

Enter Email Content/Message: Click here to verify password and claim your bank account money
Enter Embedded Link (if none, press Enter): http://bit.ly/fake-link

```

Link Security Module

HTTPS VERIFICATION

 ✓ <https://secure.com> (Safe)

 ✗ <http://secure.com> (Unsafe)

SHORT URL DETECTION

 [bit.ly](#)  Suspicious

[shorturl](#)  Suspicious

```
if (!email.getLink().empty()) {  
    if (email.getLink().find("https://") == string::npos) {  
        phishingScore += 3;  
        flags.push_back("Insecure Link (No HTTPS/SSL encryption)");  
    } else if (email.getLink().find("bit.ly") != string::npos || email.getLink().find("shorturl") != string::npos) {  
        phishingScore += 2;  
        flags.push_back("Suspicious Shortened URL detected");  
    }  
}
```

Output:

```
Enter Embedded Link (if none, press Enter): http://bit.ly/fake-link
```

Risk Scoring System

Condition	Score
Unknown Sender	+2
Phishing Keyword	+2
No HTTPS	+3
Short URL	+2

Higher score = **Higher Risk** | Scores are cumulative across all checks

Classification Logic



SAFE EMAIL

Risk Score < 3

Low Risk — Email is safe to open



PHISHING EMAIL

Risk Score ≥ 3

High Risk — Phishing detected!

Sample Input

	 Email Details
Sender:	winner@fake.com
Subject:	Free Reward Lottery
Link:	http://bit.ly/reward
 Multiple Risks	

```
Welcome to PhishShield Email Detection System
Enter Sender Email: jackpot-winner@unverified.com
Enter Subject: URGENT: You won a free reward
Enter Email Content/Message: Click here to verify password and claim your bank account money
Enter Embedded Link (if none, press Enter): http://bit.ly/fake-link
```


Sample Output



PHISHING EMAIL DETECTED

RISK LEVEL:

HIGH

DETECTED ISSUES:

- ✗ Suspicious Sender (+2)
- ✗ Phishing Keywords (+2)
- ✗ Insecure Link (+3)

----- Analysis Result -----

[ALERT] CLASSIFICATION: PHISHING EMAIL DETECTED!

Risk Level: HIGH RISK

Reasons for Warning:

- Suspicious or Unverified Sender Domain
- Contains Phishing Keyword: 'bank account'
- Contains Phishing Keyword: 'urgent'
- Contains Phishing Keyword: 'verify password'
- Contains Phishing Keyword: 'click here'
- Contains Phishing Keyword: 'free reward'
- Insecure Link (No HTTPS/SSL encryption)

Recommendation: DO NOT reply, click any links, or share personal info.

Benefits of PhishShield



Easy to use

Simple interface for quick analysis



Fast analysis

Real-time email scanning



Lightweight system

Minimal resource usage



Improves cyber awareness

Educates users about threats



Protects personal info

Safeguards sensitive data



Detects common attacks

Catches widespread phishing

Current Limitations



Rule-based detection

Relies on predefined rules, not adaptive learning



Limited trusted domains

Small whitelist may miss legitimate senders



No machine learning

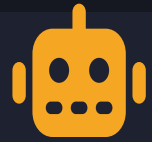
Cannot learn from new attack patterns



Cannot detect advanced phishing

Misses sophisticated social engineering

Future Improvements



AI-Based Detection

Smart pattern recognition
beyond rules



ML Integration

Train models on phishing datasets



Real-Time Monitoring

Live email stream analysis



Browser Extension

Web-based email client protection



Larger Domain DB

Expand trusted domain whitelist

Conclusion

PhishShield provides a simple and effective solution for identifying phishing emails using four core techniques:



Sender Verification

Validates sender
domain trust



Keyword Detection

Flags suspicious phrases



Link Analysis

Checks URL security



Risk Scoring

Quantifies threat level



Result: Improved Email Security



THANK YOU

